

Signal Properties

Even: $f(x) = f(-x)$

Odd: $f(x) = -f(-x)$

Periodicity:

$x(t + T) = x(t)$, where T is $\neq 0$

System Properties

For $x_1(t) \rightarrow y_1(t)$ and $x_2(t) \rightarrow y_2(t)$

The system is Linear if:

$Ax_1(t) + Bx_2(t) \rightarrow Ay_1(t) + By_2(t)$

The system is Time Invariant if:

$x_1(t + t_0) \rightarrow y_1(t + t_0)$

Time Constant:

$$T_h h(t_0) = \int_{-\infty}^{\infty} h(t) dt$$

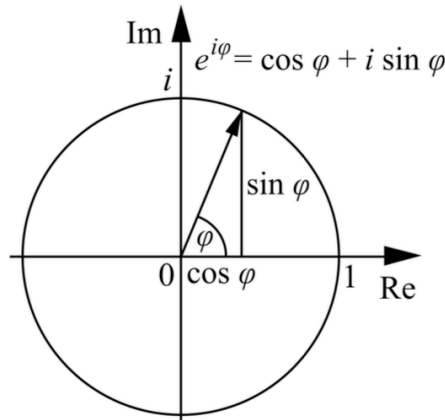
$u(t)$: Unit Step

$\delta(t)$: Unit Impulse

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

$$x_{rms} = \sqrt{P_x}$$



$$\cos x = \frac{e^{ix} + e^{-ix}}{2}$$

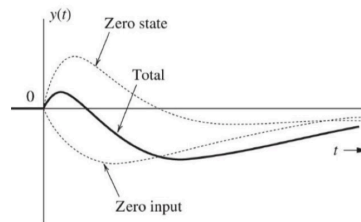
$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

$$f(t) * g(t)$$

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau$$

Linear Time Invariant Systems:

$$\text{total response} = \underbrace{\sum_{k=1}^N c_k e^{\lambda_k t}}_{\text{ZIR}} + \underbrace{x(t) * h(t)}_{\text{ZSR}}$$



The response of the system with no input applied, but (potentially) with energy stored in it at time zero.

The response of the system to an applied input signal when the system is initially at rest (i.e. IC's = 0).

NOTE: ZIR is a slightly different form when you have repeated roots.

$$Q(D)\{y(t)\} = P(D)\{x(t)\}$$

$$h(t) = [P(D)y_n(t)]u(t) \text{ when } M < N$$

With initial conditions:

$$y_n(0) = \dot{y}_n(0) = \dots = y_n^{(N-2)}(0) = 0 \text{ and } y_n^{(N-1)}(0) = 1$$

$$D^N y(t) = \frac{d^N y(t)}{dt^N}$$

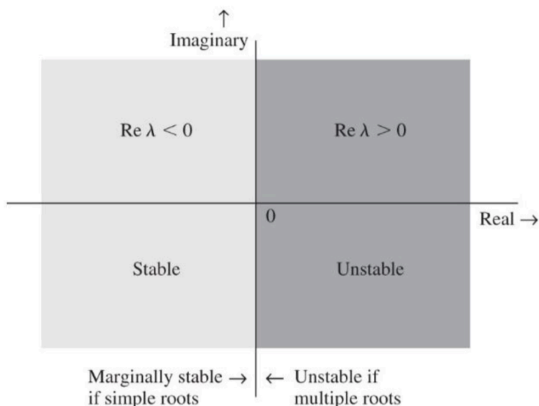


TABLE 2.1 Select Convolution Integrals

No.	$x_1(t)$	$x_2(t)$	$x_1(t) * x_2(t) = x_2(t) * x_1(t)$
1	$x(t)$	$\delta(t - T)$	$x(t - T)$
2	$e^{\lambda t} u(t)$	$u(t)$	$\frac{1 - e^{\lambda t}}{-\lambda} u(t)$
3	$u(t)$	$u(t)$	$tu(t)$
4	$e^{\lambda_1 t} u(t)$	$e^{\lambda_2 t} u(t)$	$\frac{e^{\lambda_1 t} - e^{\lambda_2 t}}{\lambda_1 - \lambda_2} u(t) \quad \lambda_1 \neq \lambda_2$
5	$e^{\lambda t} u(t)$	$e^{\lambda t} u(t)$	$te^{\lambda t} u(t)$
6	$te^{\lambda t} u(t)$	$e^{\lambda t} u(t)$	$\frac{1}{2} t^2 e^{\lambda t} u(t)$
7	$t^N u(t)$	$e^{\lambda t} u(t)$	$\frac{N! e^{\lambda t}}{\lambda^{N+1}} u(t) - \sum_{k=0}^N \frac{N! t^{N-k}}{\lambda^{k+1} (N-k)!} u(t)$
8	$t^M u(t)$	$t^N u(t)$	$\frac{M! N!}{(M+N+1)!} t^{M+N+1} u(t)$
9	$te^{\lambda_1 t} u(t)$	$e^{\lambda_2 t} u(t)$	$\frac{e^{\lambda_2 t} - e^{\lambda_1 t} + (\lambda_1 - \lambda_2) te^{\lambda_1 t}}{(\lambda_1 - \lambda_2)^2} u(t)$
10	$t^M e^{\lambda t} u(t)$	$t^N e^{\lambda t} u(t)$	$\frac{M! N!}{(N+M+1)!} t^{M+N+1} e^{\lambda t} u(t)$
11	$t^M e^{\lambda_1 t} u(t)$	$t^N e^{\lambda_2 t} u(t)$	$\sum_{k=0}^M \frac{(-1)^k M! (N+k)! t^{M-k} e^{\lambda_1 t}}{k! (M-k)! (\lambda_1 - \lambda_2)^{N+k+1}} u(t)$ $+ \sum_{k=0}^N \frac{(-1)^k N! (M+k)! t^{N-k} e^{\lambda_2 t}}{k! (N-k)! (\lambda_2 - \lambda_1)^{M+k+1}} u(t)$
	$\lambda_1 \neq \lambda_2$		
12	$e^{-\alpha t} \cos(\beta t + \theta) u(t)$	$e^{\lambda t} u(t)$	$\frac{\cos(\theta - \phi) e^{\lambda t} - e^{-\alpha t} \cos(\beta t + \theta - \phi)}{\sqrt{(\alpha + \lambda)^2 + \beta^2}} u(t)$ $\phi = \tan^{-1}[-\beta/(\alpha + \lambda)]$
13	$e^{\lambda_1 t} u(t)$	$e^{\lambda_2 t} u(-t)$	$\frac{e^{\lambda_1 t} u(t) + e^{\lambda_2 t} u(-t)}{\lambda_2 - \lambda_1} \quad \text{Re } \lambda_2 > \text{Re } \lambda_1$
14	$e^{\lambda_1 t} u(-t)$	$e^{\lambda_2 t} u(-t)$	$\frac{e^{\lambda_1 t} - e^{\lambda_2 t}}{\lambda_2 - \lambda_1} u(-t)$